

INTRODUCTION

Background:

- Methicillin-resistant *Staphylococcus aureus* (MRSA) infections remain a leading cause of antibiotic resistant infections in hospitals, and are an important cause of outpatient and emergency department (ED) visits
- National incidence of community-associated MRSA (CA-MRSA) is increasing, with three-quarters of CA-MRSA ED cases present as skin or soft tissue infection (SSTI).
- There is a need to better understand transmission pathways for this pathogen in the community-level context and to use these data to inform control strategies, especially in socioeconomically diverse and rural settings.

Objective:

- To understand community-oriented transmission pathways, we investigated spatial, social, and molecular patterns of CA-MRSA SSTI and nasal colonization in patients presenting to the UF Health Shands ED with acute SSTI

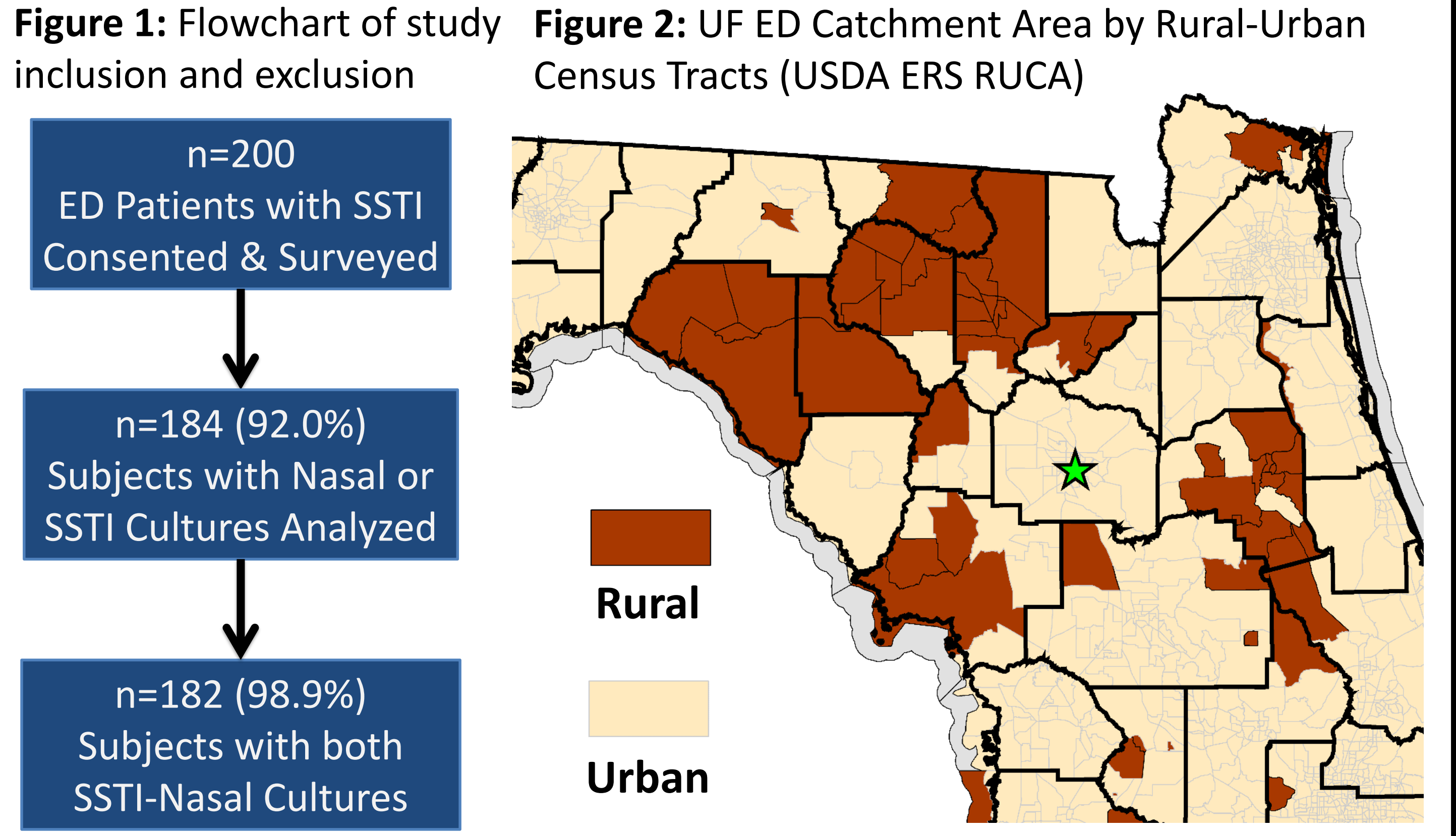
METHODS

Study Design

- From August 2015 – January 2017, pediatric and adult patients presenting to UF Health Shands Emergency Department with an acute SSTI were prospectively enrolled in a serial cross-sectional study.
- Geographic, social, and medical epidemiological data was collected via self-reported survey at time of enrollment (ED visit).
- Nasal and SSTI specimens were collected at visit and cultured for MRSA/MSSA isolation and *spa*-typing

Statistical Methods

- Self-reported home residence was geocoded and spatially joined to corresponding census tracts using ArcGIS.
- Fisher’s Exact test (nominal) and Wilcoxon rank-sum (continuous) were used to test associations between geographic, medical, and social determinants with microbiological and molecular results.
- Logistic regression was applied to assess the predictive ability of nasal colonization on SSTI culture result for MRSA and MSSA cases.
- Kernel density models (ArcMap) were developed to illustrate the spatial distribution of the study population, including MRSA and MSSA SSTI incidence



RESULTS

	SSTI Culture			Nasal Culture		
	MRSA (n=53)	Non-MRSA (n=129)	P Value	MRSA (n=25)	Non-MRSA (n=157)	P Value
Age:						
Average Age (± SD)	36.4 ± 21.0	37.8 ± 20.3	0.676	42.8 ± 19.9	36.5 ± 20.5	0.124
Pediatric (Age<18)	10 (18.9)	21 (16.3)	0.669	2 (8.0)	29 (18.5)	0.260
Sex:						
Male	31 (58.5)	66 (51.2)	0.415	13 (52.0)	84 (53.5)	0.889
Female	22 (41.5)	63 (48.8)		12 (48.0)	73 (46.5)	
Race:						
African-American	10 (18.9)	31 (24.0)	0.670	6 (24.0)	35 (22.3)	0.633
White	39 (73.6)	83 (64.3)		18 (72.0)	104 (66.2)	
Other	4 (7.5)	15 (11.7)		1 (4.0)	18 (11.5)	
Ethnicity						
Hispanic	6 (11.8)	14 (11.0)	0.920	1 (4.0)	19 (12.1)	0.318
Non-Hispanic	47 (88.2)	115 (89.0)		24 (96.0)	138 (87.9)	
Social History:						
Insurance Coverage	31 (58.5)	91 (70.5)	0.195	14 (56.0)	108 (68.8)	0.353
Established PCP	30 (56.6)	79 (61.2)	0.619	12 (48.0)	97 (61.8)	0.197
Current Alcohol Hx	21 (39.6)	44 (34.1)	0.500	14 (56.0)	51 (32.5)	0.027
Current Smoke Hx	24 (45.3)	47 (36.4)	0.316	16 (64.0)	55 (35.0)	0.008
Livestock Exposure	14 (26.4)	14 (10.9)	0.009	5 (20.0)	23 (14.7)	0.550
Antimicrobial Soap	34 (64.1)	79 (61.7)	0.629	20 (80.0)	93 (59.3)	0.071
Currently in School	6 (11.3)	23 (18.0)	0.373	1 (4.0)	28 (18.0)	0.085
Recent Incarcerated	5 (9.6)	8 (6.4)	0.668	4 (16.0)	5 (5.9)	0.221
Population Density						
<100 Per Sq. Mile	30 (56.6)	48 (37.2)	0.021	13 (52.0)	65 (41.4)	0.386
≥100 Per Sq. Mile	23 (43.4)	81 (62.8)		12 (48.0)	92 (58.6)	
Rural-Urban Status						
Rural	14 (26.4)	17 (13.1)	0.049	9 (36.0)	22 (14.0)	0.011
Urban	39 (73.6)	112 (86.9)		16 (64.0)	135 (86.0)	

CONCLUSION

- The geographic distribution of CA-MRSA is non-uniform and has a higher incidence in more rural areas and in those with livestock exposure, highlighting a possible zoonotic association in CA-MRSA transmission.
- Although previous studies have also shown the association of nasal MRSA colonization and MRSA positive wound culture, this study suggests an association of nasal MSSA colonization and MSSA positive wound cultures which has not been previously well recognized
- Associations with nasal colonization and social determinants provide insight to potential therapies to reduce CA-MRSA infections.
- Spatial clustering of CA-MRSA and MSSA SSTI exist in North-Central Florida
- Future research will assess the interactions of neighborhood-level risk factors on CA-MRSA infections as well as temporal transmission patterns across rural-urban-hospital settings using genome-wide SNP analysis.

Table 2: Adjusted* odds ratios (95%CI) of SSTI culture result as predicted by nasal colonization.

	SSTI MRSA	P Value
Nasal MRSA Colonization	34.9 (9.5 – 128.7)	<0.001

*Age, gender, race, rural-urban status

Table 3: Frequencies and distribution of MRSA SSTI *spa*-types results. MRSA isolates (n=46, 87%) were randomly selected. Includes odds ratios comparing t008 vs. other types for potential risk factor.

<i>spa</i> -type	Count (%)	t800 vs. else	OR (95%CI)
t008	35 (76.1)	Exposure vs No to Livestock	23.3 (2.2 – 246.2)
t002	2 (4.3)	Rural vs Urban Status	4.0 (0.45 – 35.5)
t1188	2 (4.3)	<100 vs ≥100 Pop. Density	1.8 (0.5 – 7.1)
t004	1 (2.2)	No vs. Recent Incarceration	12.4 (1.1 – 135.2)
t062	1 (2.2)		
t064	1 (2.2)		
t121	1 (2.2)		
t334	1 (2.2)		
t2558	1 (2.2)		
t2743	1 (2.2)		

Figure 3: Kernel density models* illustrating spatial patterns of (A) all eligible subjects (B) MRSA SSTI (C) MSSA SSTI incidence by self-reported residence. *Adjusted by neighborhood ED utilization rates

